

Are robots vegan? Unexpected behaviours in child-robot interactions and their design implications

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Figure 1: Interaction between children and humanoid robot NAO, 2022

ABSTRACT

A robot's unexpected behaviors, such as a social faux pas or system errors, affect how a child perceives or interacts with the robot. In this study, we conducted two child-robot interaction workshops on active reading in a museum of modern art, and observed the behavior and attitudes of 18 children from two age groups (6-7 yrs and 10-12 yrs). The video and audio data from this event was analyzed to observe how children in a group respond to the robot's unexpected behaviors. We extracted six different types of robot's surprising behaviors: robot's personality, movement malfunctions,

inconsistent behavior, mispronunciation, delays and freezing. We analyzed how children in the younger and the older age groups respond to each of these behaviours, and what are the similarities and differences between the two groups. Based on this analysis, we suggest guidelines for designing age-appropriate learning interactions with social robots.

CCS CONCEPTS

• Social and professional topics → Children.

KEYWORDS

Child-Robot Interaction, Social Robots, Robot Unexpected Behaviors, Child-Robot Interaction Design, Age-Appropriate Learning Experience

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1 INTRODUCTION

As social robots are becoming more and more commonplace, researchers have been exploring how they might be used with children, for example as playmates, caregivers, educators, and therapists [13] [10] [5]. In this context, it is important to understand children's attitudes towards social robots, and several studies have focused on this [19] [15] [2]. Wang et al. [16] point out the necessity of examining how to design age-appropriate interactions between children and artificial intelligence, since children have different needs at different stages of their development. One aspect of this interaction is children's attitudes towards a robot that displays unexpected behavior during the interaction [16].

Unexpected situations in child-robot interactions are usually related to the robot's appearance, behavior and system crashes [12]. Sometimes surprising situations may influence children in a positive way. Lemaignan et al. [9] found that a misbehaving robot seems to be more interesting to children compared to when it behaves as expected. Yadollahi et al. [17] suggest that robot's mistakes can have a positive influence on the child's participation in the interaction, as the child can learn through correcting the robot's inaccuracies. Of course, there are situations where a robot's unexpected behavior may not lead to a positive outcome. For example, error-prone behaviors may lower the child's trust towards the robot [8].

Sometimes the surprising behavior of a robot may have a negative effect on the child, for example by generating a feeling of fear in the child [11]. Children display certain fears related to robot's features and possible behaviors. They state uncertainty concerning robot's deceptive behaviors, possible lack of control and unpredictability of its actions [17]. According to Rubegni et al. [11], children state fears related to robot's unpredictable emotional reactions, breakdowns that can harm people, or unintentional strength lacking control. Brink et al. [3] observed that the fear of robots may come with age, where children younger than 9 years old do not display fear in relation to such technologies, but older children do.

We present here the results of our research on the design of age-appropriate learning experiences with social robots for children. We aim to answer the following research question: *how to design age-appropriate learning experiences for children in natural settings, with social robots taking into consideration children's positive and negative reactions to the robot's unexpected behaviors?* Our findings are based on a study with 18 children between 6-12 years old. We discuss the influence of robot's unexpected behavior on children's responses in the context of child-robot interaction and child's feeling of safety and propose design implications of this influence.

2 STUDY DESIGN

2.1 In-the-wild methodology

The study was conducted following the in-the-wild methodology, where a phenomenon is studied in its natural environment instead of in a laboratory. To design better real-life interactions between children and robots in the future, it is necessary to examine children's perceptions of robots in their natural environment while

they are engaged in a familiar task. Studies conducted in laboratories lack major social and contextual factors influencing children's behavior while interacting with a robot [4].

2.2 Setup of the study

Our study lasted over two days and took place in the Museum of Contemporary Art in Cracow (Poland). Two groups of children participated in this study, one each day. The first group included 8 children (6-7 yrs), and the second group included 10 children (10-12 yrs). All the children were familiar with the venue of the study as they attended some extracurricular activities at this museum.

The study was divided into two parts for both groups. The first part was a group interaction with the robot, where children could talk with the robot, ask questions, read a book together, and play a game of charades. During this part, the robot was controlled using the Wizard-of-Oz methodology. In the second part, there was a craft activity, where children were asked to draw their favorite moment from the interaction and create a book for the robot. After the activities, semi-structured interviews with the children were conducted to get their feedback on the interaction.

3 DATA COLLECTION AND ANALYSIS

Both parts of the experiment were recorded using three cameras placed at different angles and a portable microphone. A qualitative analysis was conducted using both the theme categorization and the critical incident technique. These methods were used to get an understanding of the data in terms of individual actions of the children, as well as from a broader perspective that can be grasped in a natural setting. The coding of recurring themes in the children's behavior was performed using MAXQDA software. The categories included body language (gaze direction, freezing in their seats and moving away), facial (surprise), and verbal expressions (surprise, disappointment and amazement). Instances of robot errors (interrupting a human interlocutor, mispronunciation and inconsistent answers) and malfunctions (no response, delayed response) were also coded. The critical incident technique consisted of highlighting instances of unexpected robot behavior that evoked a significant surprise or a reaction of disappointment.

4 PRELIMINARY RESULTS

A qualitative analysis of the video material and interviews from the events led to classifying unexpected robot behaviors into six different categories: expressions of robot's personality, limitations of robot's movement, inconsistent behavior, mispronunciation, delays, and freezing. Observations are structured according to two age groups: 6-7 yrs old and 10-12 yrs old. The list of all robot's unexpected behaviors observed during the study is visible in the tables located in the appendix.

Children exhibited different reactions to the robot errors. As described by Trunkle et al. [14], "Children display perseverance in their efforts to communicate with the robots, including finding ways to explain and excuse the robots' failures to communicate with them." Here we present a description of observed perseverance styles in both age groups.

4.1 Children 10-12 yrs

4.1.1 Robot's personality. During the interaction children were presented with some aspects of robot's personal profile, including its age, food preferences or knowledge. Those expressions were mainly answers to children's questions toward the robot and were mostly spontaneously created by the operators.

Older children were interested in robot's age:

Child: 'How old are you?'

NAO: 'I'm six years old.'

Older children reacted by laughing and commenting that the robot is 'so young!'. Robot's age was repeatedly mentioned later during the interviews. Children were relating it to the robot's appearance. One of them suggested that 'NAO is very short for his age'.

Children also asked robot food-related questions. They started a conversation about hot dogs:

Child: 'Do you like hot dogs?'

NAO: 'Yes, but vegan.'

Child: 'Are you vegan?'

NAO: 'Yes.'

Child: 'Why are you vegan?'

NAO: 'I love animals.'

Child: 'I love meat.'

Children reacted with surprised facial expressions and continued to ask more questions concerning this issue. They mentioned this situation later during the interviews by asking the experimenter 'Why is he vegan?'

Older children often questioned the robot's knowledge. For example, one child commented:

Child: 'In my opinion robots are stupid until you tell them what to do.'

Another child sarcastically replied to robot's words saying, 'Oh NAO, you're so smart' and started laughing.

4.1.2 Movement limitations and malfunctions. During the interactions robot faced malfunctions related to walking or moving its body parts.

At the beginning of the interaction the robot did not move when being invited to come in front of the children. The children started looking at each other and repeatedly said 'He's just shy.' Several times during the interaction, children asked the robot to perform certain movements:

Child1: 'NAO can you dance?'

Child2: 'Can you walk?'

Child3: 'NAO can you jump? NAO, jump!'

Child4: 'Do a back flip!'

Children reacted with smiles and applause to the robot sitting (from a standing posture) or dancing. They continued to ask questions about movements even after the robot stated it could not execute some of them. However, robot's evasive answers to questions challenging its movement abilities led to taunting comments from older children. Later in the interviews the children mentioned disappointment related to robot's movement abilities and that they would like it to do more 'tricks'.

4.1.3 Inconsistent behaviors. Older children reacted with surprise and confusion to inconsistencies in the robot's answers:

Child: 'He told us he can speak other languages but actually he can't. I asked him to say 'hello' in German. At first, he said he knew German. Then he said he didn't know German.'

Children also reacted with surprise to inconsistent behaviors of the robot, such as correctly guessing a child's posture during the game of charades while looking away from the child. The children reacted with surprised facial expressions, hand gestures and pointing out this incident to each other.

4.1.4 Mispronunciation. The robot mispronouncing a word often during the interaction led to confused facial expressions or further questions to the robot or the human facilitator.

Child1: 'What?'

Child2: 'Can you repeat?'

Older children exchanged taunting comments about the robot's abilities referring to those situations.

4.1.5 Delays. During the interactions sometimes there were delays in the robot answering a question. (These were caused by the slow execution of the robot's software and slow typing speed of the operator.) Often, children were asking many questions at the same time which resulted in the robot answering the previous question after a new one was asked.

Older children reacted with surprise, taunting comments or suspicious looks.

Child: 'I asked him if he could take over the world and he kind of just nodded.'

So, I was kind of just terrified, that's why I made a boogie-bomb.'

...

Child: 'Then he said he's vegan. But then I asked him <do you eat meat?> and he said <Yea>.'

This is sus (suspicious).'

4.1.6 Freezing. During the interactions the robot's software crashed a few times resulting in the robot freezing its movement and expression.

During the interaction with the older children, the robot froze at the beginning of the event when it was still behind the curtain. One of the children shouted to the robot:

Child: 'Come out, Satan!'

Later the child explained its behavior during the interview.

Child: 'He was behind the curtain, we saw just his eyes, you know, the blue circles. It was staring at us for five minutes. We found it a bit creepy.'

4.1.7 Perseverance style: nurturing. The perseverance style of nurturing was characterized as a caring and protective attitude towards the robot. The children treated the robot as if it were a baby, expressing a desire to help it learn and grow. They explained its faulty behavior in a gentle and understanding way, suggesting that the robot was simply tired or needed more time to learn.

Child: I have to teach him how to run.

Experimenter: It wasn't cool that he couldn't walk?

Child: He can walk, he just doesn't want to.

They sat close to the robot, touching it gently and speaking to it softly, as if to allow it for a better understanding of the situation. This behavior reflects a nurturing and caring approach to using technology, where the focus is on building a relationship and connection with the robot, rather than simply using it as a tool. The nurturing style was mainly expressed by female participants.

4.1.8 Perseverance style: instrumentalization. The perseverance style of instrumentalization, was characterized by a determined effort to use the robot as a tool to achieve their goals, despite its limitations. The children treated the robot as a familiar technology, similar to the voice assistants they use in their daily lives. They interacted with the robot one at a time, taking turns to speak and patiently waiting for a response. They were persistent in their efforts to communicate with the robot, repeating their questions and using simplified language to ensure that it understood them. This behavior reflects a pragmatic and goal-oriented approach to using technology, where the focus is on getting things done efficiently and effectively. The children demonstrated a willingness to work with the limitations of the robot, rather than giving up or becoming frustrated when it did not respond as expected. Overall, the perseverance style of instrumentalization is characterized by a patient and persistent effort to use technology to achieve desired outcomes.

4.1.9 Perseverance style: scepticism. The perseverance style of scepticism was characterized by a persistent and questioning attitude towards the robot. When the robot made mistakes, the children grew skeptical of its abilities and reliability. They responded by making jokes about the robot, poking fun at its errors and limitations.

However, despite their skepticism, the children did not stop interacting with the robot. Instead, they continued to talk to it while maintaining a critical and questioning attitude towards its capabilities. The children demonstrated a willingness to challenge the robot's authority and question its abilities, while still interacting with it.

4.2 Children 6-7 yrs

4.2.1 Robot's personality. Similar to the older children, younger children were also presented with some (artificial) facets of the robot's personality, including its family and friends, food preferences, or knowledge.

Younger children were more interested in robot's family and close friends:

Child1: 'Do you have friends?'
 Child2: 'Who is your favorite friend?'
 Child3: 'So do you have a friend?'
 NAO: 'I do.'

Children reacted with smiles to the information that NAO has many friends. Later, during the crafting activity, one child created a book for NAO with 'mom and dad' drawing so NAO could 'recall his own parents'.

Younger children also engaged in conversations about food with the robot:

Child: 'What do you eat for supper?'
 NAO: 'Disks.'
 Child: 'And what do you eat for second breakfast?'

NAO: 'Batteries.'

To the information about batteries as robot's favorite breakfast they reacted with smiles and repeating robot's words to each other.

Children reacted to robot's lack of knowledge about humans.

NAO: 'What do people eat?'

Child: 'NAO doesn't know everything! Because he doesn't know what people eat!'

The robot's statement led to surprise on children's faces.

4.2.2 Movement limitations and malfunctions. Younger children also observed robot malfunctions concerning walking or moving its body parts. These children explained the lack of movements with the robot's intentionality or cognitive states like 'hurting'.

Child1: 'NAO, do a step forward!'

Child2: 'He won't make it because his legs hurt.'

Child3: 'Or he doesn't want to!'

4.2.3 Inconsistent behaviors. Younger children also reacted to inconsistencies in the robot's answers. Once the robot pronounced word 'pictures' in a similar way as 'kittens':

Facilitator: 'Come on, lets make some pictures with NAO!'

NAO: 'Pictures! Pictures!'

Child: '<Kittens, kittens>?'

Children reacted with surprised facial expressions to this event.

4.2.4 Mispronunciation. Similar to the older children, younger children reacted to robot mispronouncing words with confused facial expressions, or by directing further questions toward the robot or the human facilitator. One such example is presented above under 'inconsistent behavior'. In contrast to the older children, younger children did not make taunting comments about the robot's abilities.

4.2.5 Delays. Differently from the older group, the younger children did not comment on the delay in robot's answers. They usually reacted with looking away from the robot, starting to wiggle in their seats or moving away from it.

4.2.6 Freezing. Younger children reacted to the robot's freezing by looking away from the robot, starting to wiggle in their seats, or starting to run around. During one of the freezes, they asked about taking a break for snack:

Child: 'Can we eat something now?'

4.2.7 Perseverance style: nurturing. The perseverance style of nurturing was also observed among younger children. In contrast to the older group, both boys and girls exhibited this style of behavior among the younger participants.

5 DISCUSSION AND CONCLUSIONS

5.1 Influence of unexpected behavior on child-robot interaction

Some unexpected robot behaviors positively influence child-robot interactions, such as robot personality traits, movement limitations, inconsistent behaviors, and mispronunciation. These increase children's interest, evidenced by behaviors like leaning towards the robot, asking questions, and discussing incidents. However, some

unexpected behaviors have negative impacts, such as robot personality traits related to knowledge and movement limitations. These led to skepticism from older children and boredom from younger children. These findings support previous research by Lemaignan et al. [9] and contribute novel insights on how specific forms of robot behavior influence children's behavior and perception of animacy and intelligence.

5.2 Influence of robot's unexpected behavior on child's feeling of safety

We found that inconsistent behaviors from the robot, such as delayed responses, prolonged gaze, and freezing, generated fear and feelings of unsafe interaction in older children (10-12 yrs), but not in younger children (6-7 yrs). This supports previous research [11, 18] suggesting children may fear robots due to their appearance, unpredictability, and lack of control or transparency. Our study further demonstrates that children themselves draw attention to these feelings during natural interactions. Moreover, our study found that Polish children, who have not been previously studied, also experience fear and an unsafe environment when interacting with robots. To improve child-robot interactions, designers should consider avoiding inconsistent behaviors and addressing children's fears and concerns.

5.3 Implications for designing age-appropriate child-robot interactions

5.3.1 Examining perseverance styles across different age groups. We observed different perseverance styles among children across two age groups, which might result from their different levels of cognitive and cultural development. Older children (10-12 yrs) exhibited a range of styles, including skepticism, nurturing, and instrumentalization. These children are considered to have developed abilities of reasoning and abstract thinking, as well as understanding the behavior and perspective of others [6], which may have led to their questioning the robot's abilities and reliability, but also demonstrating a caring and protective attitude towards it. Moreover, children at this age are considered to have a capacity to consciously plan and coordinate their actions, as well as an ability to solve problems based on their knowledge of the environment [6]. This may have led to their treating the robot as a tool or instrument to achieve their goals and finding ways to work around its limitations and errors. In contrast, younger children (6-7 yrs) are just starting to develop their abilities of reasoning and abstract thinking, and base their behavior largely on what they have learned from their parent-child relations [6], which results in exhibiting the perseverance style of nurturing, and treating the robot as a baby. However, some younger children were also shy to interact with the robot, perhaps due to unfamiliarity or a lack of confidence.

5.3.2 Implications for designing of interactions appropriate for 6-7-year-old children. Some of the robot's unexpected behaviors usually considered as errors, like the inability to perform certain movements, inconsistency in replying, and mispronunciation of words, can increase interest in the interaction of children of 6-7 yrs and don't have to be avoided by designers, if they appear rarely.

However, too high a frequency of robot's malfunctions, such as delays in response or freezing, could lead to boredom and distractions, and eventually children losing interest in the interaction. Such events lowering the participation rate in class activities should be taken into consideration.

Another suggestion for the designers is to create a personal profile of the robot, which may increase the children's interest in participating in class activities involving the robot. Younger children seem to be especially interested in the robot's personal life and background.

5.3.3 Implications for designing interactions appropriate for 10-12-year-old children. Our study found that in a school environment, errors that undermine the robot's knowledge or cause fear may hinder smooth child-robot interactions, which could affect children's participation in social experiences and knowledge acquisition.

To design effective child-robot interactions, designers should take care to avoid behaviors that evoke suspicion or fear. We identified two factors that may cause uneasy feelings in children: prolonged gazing and distrustful comments. To avoid prolonged gazing, designers should follow a gaze aversion pattern similar to that of human-human conversations [1]. To avoid distrustful comments, designers may develop a quick replying system to avoid delays in answering and eventual misunderstandings.

We also found that older children positively react to robots with developed personalities, which could be used to diversify school activities with robots. However, designers should avoid errors like mispronunciation of words, constant movement malfunctions, or inconsistent answers, which may provoke skepticism towards the robot's abilities or intelligence in older children.

5.4 Limitations and future research

This study was conducted in Poland with mostly Polish children attending a private international school or extracurricular activities in a museum. Those groups of children come from a richer socioeconomic background. Future research needs to explore the validity of design implications in other socioeconomic and cultural contexts, as people from different cultures react differently to robots.[7].

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Unexpected behaviors observed during the study Children of 10-12 years old	
Category of behavior	Event
Robot's personality	Robot presents information about it's age, by saying "I'm six years old."
	Robot presents information about it's food preferences, by saying that it likes vegan hotdogs and that it's vegan.
	Robot presents its knowledge by answering children's questions.
	Robot presents information about its belonging by saying "I am an owner to myself."
	Robot presents information about its origin by saying that it's from Japan.
	Robot presents information about its gender by saying that it's a boy.
Movement malfunctions	Robot can't walk while being invited to meet the children at the beginning of the interaction and has to be carried by the facilitator.
	Robot can't turn around and asks a facilitator to turn it around multiple times during the interaction.
	Robot can't dance a certain dance while being asked to.
	Robot can't run while being asked to.
	Robot can't jump while being asked to.
Inconsistent behaviors	Robot tells children it knows German language and later it tells children it doesn't know German language.
	Robot is correctly guessing child's pose during the game of charades while having it's face pointed in opposite direction than the child.
	Robot tells children that it's vegan and later on nods while a child asks it if it eats meat.
	Robot says it will dance and doesn't dance.
	Robot says "I will in a minoote" instead of "a minute".
Mispronunciation	Robot reacts to a question "Why are you vegan?" by saying "I love animals", which to children sounds as "I love lemons".

Delays	Robot provides answers after relatively long time from the questions being asked (relating to human answering questions).
	Robot answers a previous question by saying “yes” while a child asks a new one: “Do you eat meat?”
	Robot answers a previous question by nodding while a child asks a new one: “Can you take over the word?”
Freezing	Robot freezes multiple times during the interaction and doesn’t move or speak.
	Robot freezes multiple times during the interaction and doesn’t respond to children’s or facilitator’s questions.

Unexpected behaviors observed during the study
Children of 6-7 years old

Category of behavior	Event
Robot’s personality	Robot presents information about its close relations, by saying that it has friends or that it doesn’t remember its parents.
	Robot presents information about its food preferences by saying that it eats disks for supper and batteries for breakfast.
	Robot presents information about its lack of knowledge by asking children questions.
	Robot presents information about its origin by saying that it’s from Japan.
Movement malfunctions	Robot can’t walk multiple times during the interaction.
	Robot can’t move its hands when asked to.
	Robot can’t turn around and asks a facilitator to turn it around multiple times during the interaction.
Inconsistent behaviors	Robot reacts to the facilitator’s invitation to take pictures by saying “Pictures”, which to children sounds like “Kittens”.
Mispronunciation	Robot reacts to the facilitator’s invitation to take pictures by saying “Pictures”, which to children sounds like “Kittens”.

Delays	Robot provides answers after relatively long time from the questions being asked (relating to human answering questions).
Freezing	Robot freezes multiple times during the interaction and doesn’t move or speak.
	Robot freezes multiple times during the interaction and doesn’t respond to children’s or facilitator’s questions.